Final Test Coverage Report

DDP-087 • Coverage evidence and verification assessment

Darshana Opticals — Optical Shop Management System (OSMS)

DDP-087 | Evidence date: 25 October 2026 | Document status: Evidence-backed draft pending final SRS/CI traceability

Document control: Prepared from locally executed Jest/Vitest logs and DDP-087 issue requirements. No manual or E2E outcomes are invented.

1. Executive summary

Two successful local automated test executions were independently recorded: 2,075/2,075 backend tests and 1,341/1,341 frontend tests, totaling 3,416 passing tests across 228 test files/suites. Both measured repositories exceed 80% in each overall code-coverage metric. This is evidence of automated suite health at the tested local revisions, not proof that every approved SRS requirement, API endpoint, release commit, or hosted workflow has been fully verified.

Measure	Backend (Jest)	Frontend (Vitest)
Passing tests	2,075 / 2,075	1,341 / 1,341
Passing suites/files	118 / 118	110 / 110
Failed tests	0	0
Statements	92.34%	88.05%
Branches	81.68%	81.80%
Functions	97.86%	90.95%
Lines	93.00%	88.86%

2. Scope, baseline and evidence provenance

In scope: local Jest backend unit/integration tests, local Vitest/React Testing Library frontend unit/component tests, associated instrumentation and measured code coverage, plus a traceability framework for integration, browser E2E, security, performance, and requirements testing. Out of scope for these two execution logs: independently validated production readiness, hosted payment settlement, exhaustive manual UAT, and full API-to-test or SRS-to-test mapping.

Item	Backend	Frontend
Repository	darshana-opticals-osms/backend	darshana-opticals-osms/frontend
Local tested SHA (previously collected)	47eb042 (reconfirm with git rev-parse HEAD)	3e6fa9b (reconfirm with git rev-parse HEAD)
Local branch (previously collected)	docs/91-post-deployment-security-verification	test/91-owasp-security-verification
Framework	Jest; Supertest where used	Vitest 5.0.5; React Testing Library
Test execution date	25 October 2026	25 October 2026
Recorded duration	265.944 seconds	78.23 seconds
Result artifact	backend/coverage/coverage-summary.json	frontend/coverage/coverage-summary.json

Revision note: the SHAs above came from earlier local branch inspection rather than being printed inside the uploaded final coverage logs. Confirm them against git rev-parse HEAD before labeling the documents as final release evidence. The logs themselves establish the test counts and percentages.

3. Reproduction and environment

Backend command: npx jest --runInBand --testTimeout=30000 --coverage --coverageReporters=json-summary --coverageReporters=text-summary. MongoDB Memory Server used the installed MongoDB 8.3.11 executable through session-only MONGOMS_SYSTEM_BINARY; SYSTEM_BINARY_VERSION_CHECK=false and RUNTIME_DOWNLOAD=false. A separate 3/3 database foundation test passed before the full run. The earlier failed run was affected by a stuck MongoDB 8.2.6 binary download; it must not be counted as code regression evidence.

Frontend command: npm run test:coverage -- --coverage.reporter=text --coverage.reporter=json-summary. V8 was the reported coverage provider. Frontend console emitted non-failing React Router future-flag warnings and React act(...) warnings. Record these as maintenance/testing hygiene observations.

4. Coverage target assessment

Guideline requirement	Evidence	Assessment
80% code coverage for new code	All four measured overall metrics >80% in each repository	Overall threshold exceeded; new-code-only coverage not separately measured
90% branch coverage for critical business logic	Overall branch coverage: backend 81.68%; frontend 81.80%	NOT VERIFIED by aggregate numbers; per-critical-module review needed
At least one integration test per API endpoint	Integration tests present; full route inventory/test mapping unavailable	NOT VERIFIED
Critical end-to-end workflows	DDP-084 provides separate execution evidence	Must cross-check actual scenarios, URLs and statuses
Final SRS requirement traceability	Approved SRS requirement inventory not included in execution logs	NOT VERIFIED

5. Backend automated test coverage

Jest reported 118 passed suites and 2,075 passed tests. Overall statement coverage: 6,800/7,364; branch coverage: 4,958/6,070; function coverage: 778/795; line coverage: 6,571/7,065. The additional JSON field branchesTrue=100% is not equivalent to ordinary branch coverage and is not substituted for 81.68%.

Representative observed passing backend test suites include: database foundation; health; admin staff management; staff/doctor lifecycle security; chatbot session API; inventory image lifecycle/upload; loyalty read API; password reset service; prescription service; payment validation/model/monitoring; audit services; reports; authorization middleware; orders, checkout and inventory services. The list establishes observed suite-level execution, not endpoint-by-endpoint integration coverage.

6. Frontend automated test coverage

Vitest reported 110 passing test files, 1,341 passing tests, no failures. Overall coverage is 88.05% statements, 81.80% branches, 90.95% functions and 88.86% lines.

Representative executed areas: sign-up, login/password recovery, protected routes and role navigation; customer profile; product search and detail; cart/checkout/payment entry and result; order history and staff order controls; appointments/doctor sessions; prescriptions; inventory/reorder alerts; loyalty; chatbot; sales/inventory reporting; administrative staff/doctor management. Most frontend tests mock API behavior and should not be conflated with full-stack E2E validation.

7. Module-level risk observations

Frontend module	Observed branch %	Recommended interpretation
Appointment booking page	71.55%	Add boundary/error-state tests if considered critical
Payment entry page	68.70%	Critical payment initiation paths need branch review
Checkout entry page	68.05%	Checkout handling warrants targeted test review
Admin staff editor	66.37%	Sensitive role/reassignment flows warrant review
Staff move appointment page	57.64%	Uncovered move/conflict paths warrant review
Protected routes	100.00%	Strong module result; does not replace server authorization
Payment return page	100.00%	Tested component result only; hosted gateway not proven

These are file-level branch percentages from the frontend console report. The designation “critical” must be finalized by the team against SRS risk categories; no 90% compliance claim is made for them. Backend critical-file percentages require its detailed JSON file, not only the aggregate summary.

8. API integration coverage assessment

The backend suite contains integration tests, but the 118/118 result alone cannot establish one test per distinct HTTP endpoint. The required API register must be generated from the implemented router definitions, then paired with matching integration test names and methods. Authentication, permitted roles, negative-access checks and execution status must be confirmed per route. Until that comparison is completed, endpoint coverage is “Not verified.”

Endpoint group	Observed test evidence	Full endpoint inventory
Authentication/profile/recovery	Jest authentication and password-reset; frontend auth flows	Pending route-by-route review
Inventory/orders/payments	Jest inventory/checkout/payment; frontend order and payment tests	Pending route-by-route review
Appointments/prescriptions	Jest appointment/prescription; frontend page flows	Pending route-by-route review
Reports/loyalty/chatbot	Jest reporting/loyalty/chatbot; frontend services and flows	Pending route-by-route review
Admin/audit/notifications	Jest admin/audit/notification; frontend admin flows	Pending route-by-route review

9. E2E, security, performance and defect evidence

DDP-084 (infra issue #11 / merged PR #19) is the source for browser-level integration and end-to-end evidence. Prior project review referenced 10 individually reported Playwright checks, including 7/7 later repeated authorization checks; retain the actual logs and exact environment before promoting these to a final verified matrix. Hosted PayHere callback/payment verification was reported as constrained by a merchant-configuration error; do not label a complete hosted purchase journey “Pass” without new proof.

DDP-085 security: reference the existing documents/security/owasp-checklist.pdf and backend issue #91, including the precise tested conditions, scope and any outstanding limitations. DDP-086 performance: reference documents/testing/performance-report.pdf and backend issue #92; workload, p95 latency, error rate and configured test environment must be copied from that measured report rather than inferred. DDP-092 defect closure: reference infra issue #14 and associated regression evidence; merged status alone is not equivalent to verified zero residual defects.

10. CI and release traceability

CI workflow run IDs, artifact URLs and last release SHAs are not present in the uploaded coverage logs. Record exact GitHub Actions run URLs and confirm the tested commits match the delivered release. If the local SHAs differ from final develop/main, state the difference and obtain appropriate release verification rather than silently treating local results as production sign-off.

11. Open gaps and corrective follow-up

- Confirm actual tested SHAs, target branch and release commit; attach persistent log/coverage artifacts.
- Map every SRS functional and applicable nonfunctional requirement to test cases; do not invent SRS IDs.
- Inventory every implemented backend API endpoint and document at least one matching integration test or explicit gap.
- Assess each team-designated critical business-logic module against its 90% branch target.
- Cross-check DDP-084 browser E2E and hosted payment outcome with logs; mark unresolved items Blocked/Not Verified.
- Import actual DDP-085, DDP-086 and DDP-092 measured results and traceable issues/CI evidence.

12. Conclusion and sign-off status

Conclusion: the reported backend and frontend local automated suites PASSED (3,416/3,416 tests), and both independently measured aggregate code-coverage profiles are above 80% in statements, branches, functions and lines. Separate proof remains necessary for critical-module branch targets, route completeness, full SRS traceability, E2E hosted payments and final release parity. Therefore this report supports automated-test completion at the observed local revisions, but is not an unconditional final UAT or production-readiness certificate.

Prepared for DDP-087 review. QA reviewer: _____ Review date: _____ Release SHA(s): _____